

# Lecture #4: Probability Spaces & Discrete Random Variables

MATH 145 – Calculus I

Torin Quinlivan

2025-09-22

# Review: Axioms of Probability

**Definition:** A **probability space** is a triple  $(\Omega, \mathcal{F}, P)$ , where  $\Omega$  is the **sample space**,  $\mathcal{F}$  is the **event space**, and  $P$  is a **probability law (or measure)**.

**Definition:** A **probability law** is a function  $P : \mathcal{F} \rightarrow [0, 1]$  that maps an event  $A$  to a real number in  $[0, 1]$ . The function must satisfy the **axioms of probability**:

1. **Non-Negativity:**  $P(A) \geq 0$ , for any  $A \subseteq \Omega$ .
2. **Normalization:**  $P(\Omega) = 1$ .
3. **Additivity:** For any disjoint sets  $\{A_1, A_2, A_3, \dots\}$ , it must be true that

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

# Review: Important Properties and Results

**Corollary:** Let  $A \in \mathcal{F}$  be an event. Then:

1.  $P(\overline{A}) = 1 - P[A]$ .
2.  $P(A) \leq 1$ .
3.  $P(\emptyset) = 0$ .

**Corollary (Unions of Two Non-Disjoint Sets):** For any  $A$  and  $B$  in  $\mathcal{F}$ ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

# Review: Important Properties and Results

**Corollary (Inequalities):** Let  $A$  and  $B$  be two events in  $\mathcal{F}$ . Then:

1.  $P(A \cup B) \leq P(A) + P(B)$ . (Union Bound)
2. If  $A \subseteq B$ , then  $P(A) \leq P(B)$ .

**Theorem (Bayes' Theorem):** For any two events  $A$  and  $B$  such that  $P(A) > 0$  and  $P(B) > 0$ ,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

# Review: Important Properties and Results

**Theorem (Law of Total Probability):** Let  $\{A_1, A_2, A_3, \dots\}$  be a partition of  $\Omega$ , i.e.,  $A_1, A_2, A_3, \dots$  are disjoint and  $\Omega = A_1 \cup A_2 \cup \dots$ . Then, for any  $B \subseteq \Omega$ ,

$$P(B) = \sum_{i \in I} P(B | A_i) P(A_i).$$

**Corollary:** Let  $\{A_1, A_2, A_3, \dots\}$  be a partition of  $\Omega$ , i.e.,  $A_1, A_2, A_3, \dots$  are disjoint and  $\Omega = A_1 \cup A_2 \cup \dots$ . Then, for any  $B \subseteq \Omega$ ,

$$P(A_j | B) = \frac{P(B | A_j) P(A_j)}{\sum_{i \in I} P(B | A_i) P(A_i)}.$$

# Prove You're a Probability Space!

**Example:** Let  $S$  be a (non-empty) set. Then  $(S, \mathcal{P}(S), P)$  is a probability space, where  $P(A) = \frac{|A|}{|S|}$  for any  $A \in \mathcal{P}(S)$ .

**Example:** Let  $A \subset \Omega$ , where  $(\Omega, \mathcal{F}, P)$  is a probability space and  $P(A) > 0$ . Show  $\widehat{P}(B) = \frac{P(B \cap A)}{P(A)}$  defines a probability measure on  $\mathcal{F} \cap A$ .

# Random Variables

Let  $(\Omega, \mathcal{F}, P)$  be a probability space.

**Definition:** A **random variable**  $X$  is a function  $X : \Omega \rightarrow \mathbb{R}$  that maps an outcome  $\xi \in \Omega$  to a number  $X(\xi)$  on the real line.

**Definition:** A random variable  $Y$  is said to be **discrete** if it can assume only a finite or countably infinite number of distinct values.

**Example:** Let  $\Omega$  consist of the outcomes of tossing a (fair) coin three times and let  $X : \Omega \rightarrow \mathbb{R}$  be the number of heads that appears.

# Probability Mass Functions

**Definition:** The **probability mass function (PMF)** of a random variable  $X$  is a function which specifies the probability of obtaining a number  $X(\xi) = x$ . We denote a probability mass function as:

$$p_X(x) = P(X = x).$$

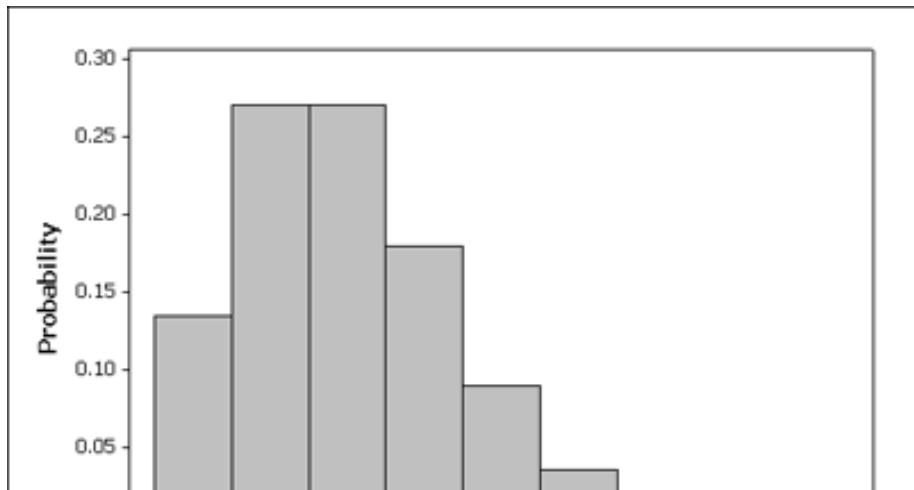
The set of all possible states of  $X$  is denoted as  $X(\Omega)$ .

**Example:** Let  $\Omega$  consist of the outcomes of tossing a (fair) coin three times and let  $X : \Omega \rightarrow \mathbb{R}$  be the number of heads that appears. What is the probability mass function (PMF) of this random variable?



# Properties of Probability Mass Functions

Random variables have a graphical representation called a **probability histogram**.



# Properties of Probability Mass Functions

**Theorem:** If  $A \subset X(\Omega)$ , then the function  $P(A) = \sum_{x \in A} p_X(x)$  is a probability law of  $X(\Omega)$ .

Can you prove this?

# The Expected Value of a Random Variable

**Definition:** Let  $X$  be a discrete random variable with the probability function  $p_X(x)$ . Then the **expected value (or mean)** of  $X$ ,  $E(X)$ , is defined to be:

$$E(X) = \sum_x x p_X(x).$$

If  $p_X(x)$  is an accurate characterization of the population frequency distribution, then  $E(X) = \mu$ , the population mean.

# The Expected Value of a Function of a Random Variable

**Definition:** Let  $X$  be a discrete random variable with probability function  $p_X(x)$  and  $g(X)$  be a real-valued function of  $X$ . Then the expected value of  $g(X)$  is given by:

$$E(g(X)) = \sum_x g(x) p_X(x).$$

# Moments of a Random Variable

A specific case of an expected value of a function of a random variables are the **moments** of the random variables.

The  $n^{th}$  **moment** of a random variable  $X$  is defined as:

$$E(X^n) = \sum_x x^n p_X(x).$$

The  $n^{th}$  **central moment** of a random variable  $X$  is defined as:

$$E(X^n) = \sum_x (x - \mu)^n p_X(x),$$

where  $\mu = E(X)$ .

# Properties of the Expected Value

**Theorem:** Let  $X$  be a discrete random variable with probability function  $p_X(x)$  and  $c$  be a constant. Then  $E(c) = c$ .

**Theorem:** Let  $X$  be a discrete random variable with probability function  $p_X(x)$ ,  $g(X)$  be a function of  $X$ , and  $c$  be a constant. Then:

$$E[c \times g(X)] = c \times E[g(X)].$$

**Theorem:** Let  $X$  be a discrete random variable with probability function  $p_X(x)$  and  $g_1(X), g_2(X), \dots, g_k(X)$  be  $k$  functions of  $X$ . Then:

$$E \left[ \sum_{i=1}^k g_i(X) \right] = \sum_{i=1}^k E[g_i(X)].$$

# Variance of a Random Variable

$E(X)$  gives the mean of the population with distribution given by  $p_X(x)$ . Now let's define the variance and standard deviation of this population.

**Definition:** The **variance** of a set of measurements is the average of the square of the differences between the values in a set of measurements and their mean, denoted as  $\sigma^2$ .

**Definition:** The **standard deviation** of  $X$  is the positive square root of  $V(X)$ , denoted as  $\sigma$ .

# Formulas for the Variance

**Definition:** If  $X$  is a random variable with mean  $\mu = E(X)$ , the **variance** of a random variable  $X$  is defined to be the expected value of  $(X - \mu)^2$ , or the second central moment. That is,

$$V(X) = E[(X - \mu)^2].$$

**Theorem:** Let  $X$  be a discrete random variable with probability function  $p_X(x)$  and finite mean  $E(X) = \mu$ ; then:

$$V(X) = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$$

This second formula is a lot easier to use!



# Properties of the Variance

**Theorem:** Let  $X$  be a discrete random variable with probability mass function  $p_X(x)$  and  $c$  be a constant. Then  $V(c) = 0$ .

**Theorem:** Let  $X$  be a discrete random variable with probability mass function  $p_X(x)$  and  $c$  be a constant. Then:

$$V[c \times X] = c^2 \times V[X].$$

**Theorem:** Let  $X_1, X_2, X_3, \dots$  be discrete random variables with probability mass functions  $p_{X_1}(x), p_{X_2}(x), p_{X_3}(x), \dots$ . Then:

$$V \left[ \sum_{i=1}^k X_i \right] = \sum_{i=1}^k V[X_i].$$